

Chang Chip Hong

professor

基本信息

姓名:	Chang Chip Hong	所属院校:	南洋理工大学
邮箱:	echchang@ntu.edu.sg	地区:	Singapore
年龄范围:	50岁以上	企业经历:	有

学科领域

信息安全 电子与计算机工程 集成电路设计与集成系统 网络空间安全

研究领域

电路与系统设计 网络安全理论与技术 数字集成电路设计 安全协议与安全认证

数据隐私保护 低功耗电路与能量管理 集成电路设计与制造 计算机体系结构

低功耗设计与能耗优化 密码学与密码分析 +5

个人简介

Professor Chang Chip Hong holds B.Eng., M.Eng., and Ph.D. degrees from the National University of Singapore and Nanyang Technological University (NTU), Singapore. He has been a faculty member at NTU's School of Electrical and Electronic Engineering since 1999 and currently serves as a Professor and Chief Consultant of Rock Solid Research Lab. His extensive experience includes roles as Assistant Chair of Alumni, Deputy Director of the Center for High Performance Embedded Systems, and Program Director of the Center for Integrated Circuits and Systems. His research focuses on hardware security, AI security, biometric security, trustworthy sensing, and hardware accelerators for post-quantum cryptography. He has authored/coedited six books, published

over 100 international journal papers, and delivered more than 50 keynotes and tutorials.

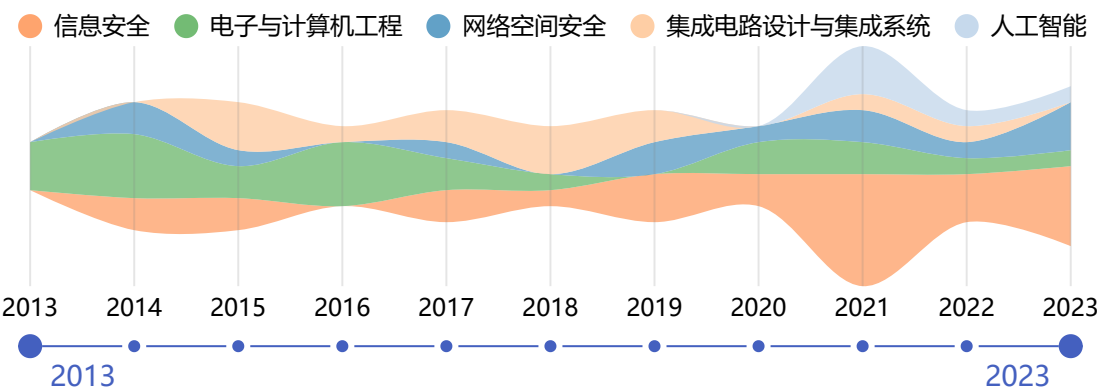
工作经历

Nanyang Technological University - Professor	2022-09-01 -
Nanyang Technological University - Associate Professor	2005-10-01 - 2022-08-31
Nanyang Technological University - Assistant Professor	1999-07-01 - 2005-09-30

教育经历

Nanyang Technological University - PhD	- 1998
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近年发表的论文



Scalable and Conflict-Free NTT Hardware Accelerator Design: Methodology, Proof, and Implementation
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS
Mu, Jianan; Ren, Yi; Wang, Wen; Hu, Yizhong; Chen, Shuai; Chang, Chip-Hong; Fan, Junfeng; Ye, Jing; Cao, Yuan; Li, Huawei; Li, Xiaowei
被引用次数: 8 2023

A New Reconfigurable True Random Number Generator and Physical Unclonable Function Unified Chip With On-Chip Auto-Calibration
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS
Cao, Yuan; Liu, Wanyi; Zheng, Yue; Chen, Shuai; Ye, Jing; Qian, Lei; Chang, Chip-Hong
被引用次数: 1 2023

Guest Editorial Special Issue on the Asian Hardware Oriented Security and Trust Symposium (AsianHOST 2022)
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS
Chang, Chip Hong; Zhou, Pingqiang; Cao, Yuan; Liu, Qiang

被引用次数: 0

2023

An Imperceptible Data Augmentation Based Blackbox Clean-Label Backdoor Attack on Deep Neural Networks

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Xu, Chaohui; Liu, Wenye; Zheng, Yue; Wang, Si; Chang, Chip-Hong

被引用次数: 3

2023

PUF-Based Mutual Authentication and Key Exchange Protocol for Peer-to-Peer IoT Applications

IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING

Zheng, Yue; Liu, Wenye; Gu, Chongyan; Chang, Chip-Hong

被引用次数: 24

2023

An Efficient Full Hardware Implementation of Extended Merkle Signature Scheme

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cao, Yuan; Wu, Yanze; Wang, Wen; Lu, Xu; Chen, Shuai; Ye, Jing; Chang, Chip-Hong

被引用次数: 7

2022

A New Energy-Efficient and High Throughput Two-Phase Multi-Bit per Cycle Ring Oscillator-Based True Random Number Generator

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cao, Yuan; Zhao, Xiaojin; Zheng, Wenhan; Zheng, Yue; Chang, Chip-Hong

被引用次数: 25

2022

A Dual-Mode Polarization Image Sensor Design Based on In-Pixel Dual-Band Metal Wire Grid for Trustworthy Sensing

IEEE SENSORS JOURNAL

Zheng, Hongxia; Zuo, Haibiao; Zhao, Xiaojin; Yang, Yatao; Chang, Chip-Hong

被引用次数: 0

2022

Radio Frequency Fingerprints vs. Physical Unclonable Functions-Are They Twins, Competitors, or Allies?

IEEE NETWORK

Zhang, Junqing; Chang, Chip-Hong; Gu, Chongyan; Hanzo, Lajos

被引用次数: 3

2022

A DNN Fingerprint for Non-Repudiable Model Ownership Identification and Piracy Detection

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Zheng, Yue; Wang, Si; Chang, Chip-Hong

被引用次数: 11

2022

Area, Time and Energy Efficient Multicore Hardware Accelerators for Extended Merkle Signature Scheme

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cao, Yuan; Wu, Yanze; Qin, Lan; Chen, Shuai; Chang, Chip-Hong

被引用次数: 1

2022

Halide perovskite memristors as flexible and reconfigurable physical unclonable functions

NATURE COMMUNICATIONS

John, Rohit Abraham; Shah, Nimesh; Vishwanath, Sujaya Kumar; Ng, Si En; Febriansyah, Benny; Jagadeeswararao, Metikoti; Chang, Chip-Hong; Basu, Arindam; Mathews, Nripan

被引用次数: 126

2021

A New Lightweight In Situ Adversarial Sample Detector for Edge Deep Neural Network

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Wang, Si; Liu, Wenye; Chang, Chip-Hong

被引用次数: 4

2021

Two Sides of the Same Coin: Boons and Banes of Machine Learning in Hardware Security

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Liu, Wenye; Chang, Chip-Hong; Wang, Xueyang; Liu, Chen; Fung, Jason M.; Ebrahimabadi, Mohammad; Karimi, Naghmeh; Meng, Xingyu; Basu, Kanad

被引用次数: 9

2021

Hardware Security in Emerging Technologies: Vulnerabilities, Attacks, and Solutions

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Karimi, Naghmeh; Basu, Kanad; Chang, Chip-Hong; Fung, Jason M.

被引用次数: 3

2021

A large-scale comprehensive evaluation of single-slice ring oscillator and PicoPUF bit cells on 28-nm Xilinx FPGAs

JOURNAL OF CRYPTOGRAPHIC ENGINEERING

Gu, Chongyan; Chang, Chip-Hong; Liu, Weiqiang; Hanley, Neil; Miskelly, Jack; O'Neill, Maire

被引用次数: 17

2021

Identification of FSM State Registers by Analytics of Scan-Dump Data

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Cui, Aijiao; He, Chengkang; Chang, Chip-Hong; Lu, Hao

被引用次数: 0

2021

Stealthy and Robust Glitch Injection Attack on Deep Learning Accelerator for Target With Variational Viewpoint

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Liu, Wenye; Chang, Chip-Hong; Zhang, Fan

被引用次数: 3

2021

A New PUF Based Lock and Key Solution for Secure In-Field Testing of Cryptographic Chips

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING

Cui, Aijiao; Chang, Chip-Hong; Zhou, Wei; Zheng, Yue

被引用次数: 29

2021

A PUF-Based Data-Device Hash for Tampered Image Detection and Source Camera Identification

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Zheng, Yue; Cao, Yuan; Chang, Chip-Hong

被引用次数: 28

2020

Ed-PUF: Event-Driven Physical Unclonable Function for Camera Authentication in Reactive Monitoring System

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Zheng, Yue; Zhao, Xiaojin; Sato, Takashi; Cao, Yuan; Chang, Chip-Hong

被引用次数: 10

2020

A New Polarization Image Demosaicking Algorithm by Exploiting Inter-Channel Correlations With Guided Filtering

IEEE TRANSACTIONS ON IMAGE PROCESSING

Liu, Shumin; Chen, Jiajia; Xun, Yuan; Zhao, Xiaojin; Chang, Chip-Hong

被引用次数: 21

2020

Emerging Attacks and Solutions for Secure Hardware in the Internet of Things

IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING

Chang, Chip Hong; van Dijk, Marten; Ruehrmair, Ulrich; Tehranipoor, Mark M.

被引用次数: 2

2019

UDhashing: Physical Unclonable Function-Based User-Device Hash for Endpoint Authentication

IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS

Zheng, Yue; Cao, Yuan; Chang, Chip-Hong

被引用次数: 20

2019

Area- and Power-Efficient Nearly-Linear Phase Response IIR Filter by Iterative Convex Optimization

IEEE ACCESS

Deng, Gelei; Chen, Jiajia; Zhang, Jiaxuan; Chang, Chip-Hong

被引用次数: 4

2019

An Energy-Efficient Current-Starved Inverter Based Strong Physical Unclonable Function With Enhanced Temperature Stability

IEEE ACCESS

Cao, Yuan; Zheng, Wenhan; Zhao, Xiaojin; Chang, Chip-Hong

被引用次数: 16

2019

Reliable and Modeling Attack Resistant Authentication of Arbiter PUF in FPGA Implementation With Ternary Quadruple Response

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Zalivaka, Siarhei S.; Ivaniuk, Alexander A.; Chang, Chip-Hong

被引用次数: 83

2019

A New Accurate and Fast Homography Computation Algorithm for Sports and Traffic Video Analysis

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Liu, Shumin; Chen, Jiajia; Chang, Chip-Hong; Ai, Ye

被引用次数: 11

2018

Tap Delay-and-Accumulate Cost Aware Coefficient Synthesis Algorithm for the Design of Area-Power Efficient FIR Filters

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Jiajia; Chang, Chip-Hong; Ding, Jiatao; Qiao, Rui; Faust, Mathias

被引用次数: 9

2018

New Hardware and Power Efficient Sporadic Logarithmic Shifters for DSP Applications

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Chen, Jiajia; Chang, Chip-Hong; Wang, Yujia; Zhao, Juan; Rahardja, Susanto

被引用次数: 4

2018

Current Mirror Array: A Novel Circuit Topology for Combining Physical Unclonable Function and Machine Learning

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wang, Zheng; Chen, Yi; Patil, Aakash; Jayabalan, Jayasanker; Zhang, Xueyong; Chang, Chip-Hong; Basu, Arindam

被引用次数: 32

2018

A Low Power Diode-Clamped Inverter-Based Strong Physical Unclonable Function for Robust and Lightweight Authentication

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cao, Yuan; Liu, Chao Qun; Chang, Chip Hong

被引用次数: 45

2018

A New Cost-Aware Sensitivity-Driven Algorithm for the Design of FIR Filters

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Jiajia; Tan, Jinghong; Chang, Chip-Hong; Feng, Feng

被引用次数: 16

2017

New Algorithm for Signed Integer Comparison in $\{2n+k, 2n-1, 2n+1, 2n\pm 1-1\}$ and Its Efficient Hardware Implementation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kumar, Sachin; Chang, Chip-Hong; Tay, Thian Fatt

被引用次数: 11

2017

ACRO-PUF: A Low-power, Reliable and Aging-Resilient Current Starved Inverter-Based Ring Oscillator Physical Unclonable Function

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Liu, Chao Qun; Cao, Yuan; Chang, Chip Hong

被引用次数: 66

2017

A Scaling-Assisted Signed Integer Comparator for the Balanced Five-Moduli Set RNS $\{2n-1, 2n, 2n+1, 2n+1-1, 2n-1-1\}$

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kumar, Sachin; Chang, Chip-Hong

被引用次数: 5

2017

Static and Dynamic Obfuscations of Scan Data Against Scan-Based Side-Channel Attacks

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Cui, Aijiao; Luo, Yanhui; Chang, Chip-Hong

被引用次数: 61

2017

A Retrospective and a Look Forward: Fifteen Years of Physical Unclonable Function Advancement

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Chang, Chip-Hong; Zheng, Yue; Zhang, Le

被引用次数: 112

2017

Asymmetrical domain wall propagation in bifurcated PMA wire structure due to the Dzyaloshinskii-Moriya interaction

APPLIED PHYSICS LETTERS

Kwon, J.; Goolaup, S.; Gan, W. L.; Chang, C. H.; Roy, K.; Lew, W. S.

被引用次数: 3

2017

Cyclic resistance change in perpendicularly magnetized Co/Ni nanowire induced by alternating current pulse injection

CURRENT APPLIED PHYSICS

Kwon, J.; Goolaup, S.; Tan, F. N.; Chang, C. H.; Roy, K.; Lew, W. S.

被引用次数: 2

2017

A New Paradigm of Common Subexpression Elimination by Unification of Addition and Subtraction

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Ding, Jiatao; Chen, Jiajia; Chang, Chip-Hong

被引用次数: 20

2016

Efficient VLSI Implementation of $2n$ Scaling of Signed Integer in RNS $\{2n-1, 2n, 2n+1\}$ (vol 21, pg 1936, 2013)

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Tay, Thian Fatt; Chang, Chip-Hong; Low, Jeremy Yung Shern

被引用次数: 0

2016

A New Fast and Area-Efficient Adder-Based Sign Detector for RNS $\{2n-1, 2n, 2n+1\}$

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kumar, Sachin; Chang, Chip-Hong

被引用次数: 18

2016

DW-AES: A Domain-Wall Nanowire-Based AES for High Throughput and Energy-Efficient Data Encryption in Non-Volatile Memory

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Wang, Yuhao; Ni, Leibin; Chang, Chip-Hong; Yu, Hao

被引用次数: 36

2016

Hypergraph Based Minimum Arborescence Algorithm for the Optimization and Reoptimization of Multiple Constant Multiplications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Feng, Feng; Chen, Jiajia; Chang, Chip-Hong

被引用次数: 14

2016

A Non-Iterative Multiple Residue Digit Error Detection and Correction Algorithm in RRNS

IEEE TRANSACTIONS ON COMPUTERS

Tay, Thian Fatt; Chang, Chip-Hong

被引用次数: 26

2016

Highly Reliable Spin-Transfer Torque Magnetic RAM-Based Physical Unclonable Function With Multi-Response-Bits Per Cell

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Zhang, Le; Fong, Xuanyao; Chang, Chip-Hong; Kong, Zhi Hui; Roy, Kaushik

被引用次数: 33

2015

A Low-Power Hybrid RO PUF With Improved Thermal Stability for Lightweight Applications

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Cao, Yuan; Zhang, Le; Chang, Chip-Hong; Chen, Shoushun

被引用次数: 116

2015

CMOS Image Sensor Based Physical Unclonable Function for Coherent Sensor-Level Authentication

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cao, Yuan; Zhang, Le; Zalivaka, Siarhei S.; Chang, Chip-Hong; Chen, Shoushun

被引用次数: 68

2015

Base Transformation With Injective Residue Mapping for Dynamic Range Reduction in RNS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Tay, Thian Fatt; Chang, Chip-Hong; Sousa, Leonel

被引用次数: 5

2015

Novel Design Algorithm for Low Complexity Programmable FIR Filters Based on Extended Double Base Number System

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Jiajia; Chang, Chip-Hong; Feng, Feng; Ding, Weiao; Ding, Jiatao

Optimizating Emerging Nonvolatile Memories for Dual-Mode Applications: Data Storage and Key Generator

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhang, Le; Fong, Xuanyao; Chang, Chip-Hong; Kong, Zhi Hui; Roy, Kaushik

被引用次数: 30

2015

Low field domain wall dynamics in artificial spin-ice basis structure

JOURNAL OF APPLIED PHYSICS

Kwon, J.; Goolaup, S.; Lim, G. J.; Kerk, I. S.; Chang, C. H.; Roy, K.; Lew, W. S.

被引用次数: 4

2015

Design of Optimal Scan Tree Based on Compact Test Patterns for Test Time Reduction

IEEE TRANSACTIONS ON COMPUTERS

Chen, Linfeng; Cui, Aijiao; Chang, Chip-Hong

被引用次数: 11

2015

A Cluster-Based Distributed Active Current Sensing Circuit for Hardware Trojan Detection

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Cao, Yuan; Chang, Chip-Hong; Chen, Shoushun

被引用次数: 47

2014

A Blind Dynamic Fingerprinting Technique for Sequential Circuit Intellectual Property Protection

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Chang, Chip-Hong; Zhang, Li

被引用次数: 29

2014

A 0.7 V low-power fully programmable Gaussian function generator for brain-inspired Gaussian correlation associative memory

NEUROCOMPUTING

Li, Fei; Chang, Chip-Hong; Basu, Arindam; Siek, Liter

被引用次数: 7

2014

Exploiting Process Variations and Programming Sensitivity of Phase Change Memory for Reconfigurable Physical Unclonable Functions

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Zhang, Le; Kong, Zhi Hui; Chang, Chip-Hong; Cabrini, Alessandro; Torelli, Guido

被引用次数: 84

2014

A Pragmatic Per-Device Licensing Scheme for Hardware IP Cores on SRAM-Based FPGAs

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Zhang, Li; Chang, Chip-Hong

被引用次数: 14

2014

A New Approach to the Design of Efficient Residue Generators for Arbitrary Moduli

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Low, Jeremy Yung Shern; Chang, Chip-Hong

被引用次数: 18

2013

Radix-4 and Radix-8 Booth Encoded Multi-Modulus Multipliers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Muralidharan, Ramya; Chang, Chip-Hong

被引用次数: 16

2013

Efficient VLSI Implementation of 2n Scaling of Signed Integer in RNS {2n-1, 2n, 2n+1}

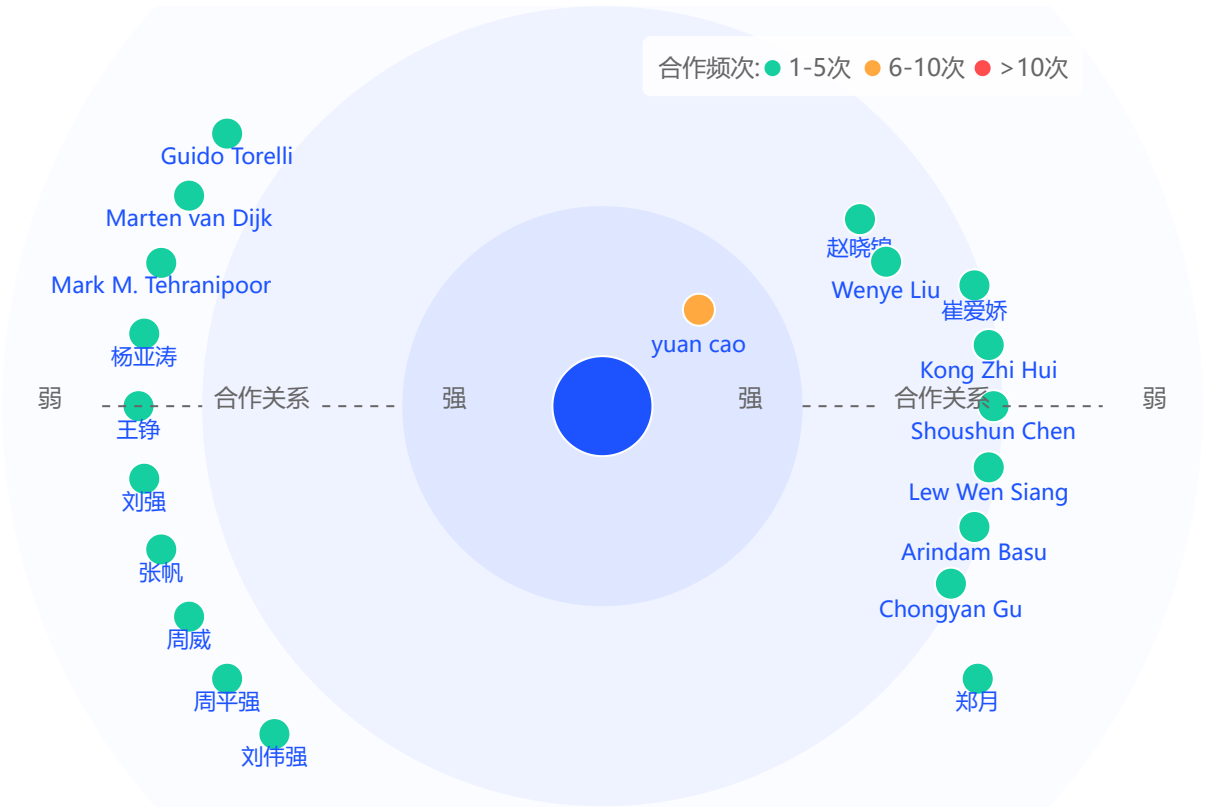
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Tay, Thian Fatt; Chang, Chip-Hong; Low, Jeremy Yung Shern

被引用次数: 18

2013

学术合作



yuan cao	合作次数: 10
南洋理工大学	
赵晓锦	合作次数: 5
深圳大学	
Wenye Liu	合作次数: 5
南洋理工大学	

崔爱娇
哈尔滨工业大学

合作次数: 3

Kong Zhi Hui
南洋理工大学

合作次数: 3

还有 30 个合作者